

1 Introduction to MATLAB

1.1 MATLAB Environment

- MATLAB and its role in Chemical Engineering
- MATLAB Desktop and MATLAB Online
- Command Window, Workspace, Current Folder
- Editor, Command History
- Help and documentation

1.2 Getting Started

- Basic calculations
- Variables and assignment
- Constants and built-in functions
- Scripts and Live Scripts
- Saving and running MATLAB programs

2 MATLAB Data Types and Data Structures

2.1 Fundamental Data Types

- Numeric data: `double`, `single`
- Logical data
- Character arrays and Strings

2.2 Date and Time Data

- `datetime` and `duration` types
- Basic date/time calculations
- Formatting dates and durations

2.3 MATLAB Data Structures

- Cell arrays, Structure arrays, Table arrays
- Accessing and modifying structured data

2.4 Specialized Data Types

- Sparse arrays and when to use them
- Basic creation and manipulation

2.5 Data Type Conversion

- Numeric, string, and logical conversion
- Converting between tables, cells, and structures
- Checking data types with `class`, `isa`

3 Vectors, Matrices, and Array Operations

3.1 Vectors and Matrices

- Scalars, row vectors, column vectors
- Matrix creation and multidimensional arrays
- Array initialization

3.2 Indexing and Manipulation

- Array indexing and logical indexing
- Slicing, concatenation, reshaping
- Transpose operations

3.3 Matrix Mathematics

- Matrix addition and multiplication
- Element-wise operations
- Matrix division, inverse, determinant
- Matrix rank and gradient

4 MATLAB Operators and Logic

4.1 Operators

- Arithmetic, relational, and logical operators
- Operator precedence
- Matrix vs. element-wise operations

4.2 Logical Programming

- AND, OR, NOT, XOR, XNOR operations
- Truth tables
- Logical indexing and engineering applications

5 Control Structures

5.1 Conditional Statements

- if, elseif, else
- Nested conditions

5.2 Selection Structures

- switch, case, otherwise

5.3 Loops

- `for`, `while` loops
- Nested loops
- `break` and `continue` statements

5.4 Engineering Applications

- Iterative calculations
- Conditional process calculations
- Repetitive engineering computations

6 Functions and Modular Programming

6.1 Introduction to Functions

- Purpose, syntax, inputs, and outputs
- Local variables and function scope

6.2 Types of MATLAB Functions

- Inline functions and anonymous functions
- `.m` function files and subfunctions
- Function handles

6.3 Variable Scope and Function Memory

- Local and global variables
- Persistent variables with `persistent`

6.4 Practical Function Development

- Multiple inputs and outputs
- Default behavior and argument checking
- Reusable engineering functions
- Organizing MATLAB projects

7 Random Number Generation and Basic Statistical Analysis

7.1 Random Numbers

- Random number generation: `rand`, `randi`, `randn`
- Random number types and distributions
- Reproducibility with random seeds

7.2 Descriptive Statistics

- Mean, median, variance, standard deviation
- Range and coefficient of variation

7.3 Regression and Correlation

- Covariance and correlation coefficient
- Correlation tests and statistical significance
- Coefficient of determination (R^2)

7.4 Engineering Data Analysis

- Experimental data and measurement variability
- Comparing datasets
- Basic statistical interpretation

8 Data Visualization

8.1 Basic Plots

- Line plots, scatter plots, bar plots
- Histograms and boxplots

8.2 Engineering Data Visualization

- Discrete data plotting
- Multiple datasets and curves
- Subplots and error bars

8.3 Advanced Axis Options

- Logarithmic axes (semi-log and log-log plots)
- Axis scaling and limits

8.4 Plot Customization

- Titles, axis labels, units
- Legends, annotations, grid
- Exporting figures

9 File Handling and Input/Output

9.1 File Management

- File paths and directories
- File operations: `fopen`, `fclose`, `exist`
- File existence checking

9.2 Reading and Writing Data

- Formatted input/output: `fprintf`, `fscanf`
- Reading and writing text files
- CSV and structured data

9.3 File Position and Error Handling

- File pointers and positions: `ftell`, `feof`
- Detecting end-of-file
- Handling file errors with `ferror`

9.4 Engineering Data Import/Export

- Working with experimental datasets
- Saving MATLAB data
- Exporting calculated results

10 Symbolic Mathematics

10.1 Symbolic Variables and Expressions

- Symbolic variables and constants
- Symbolic expressions

10.2 Symbolic Algebra

- Simplification, expansion, factorization
- Substitution operations

10.3 Symbolic Equations

- Solving equations and systems of equations
- Using `solve`

10.4 Calculus

- Symbolic differentiation and integration
- Limits

10.5 Chemical Engineering Applications

- Analytical material balances
- Rate expressions
- Analytical integration
- Symbolic derivation of engineering equations

11 Numerical Solution of Algebraic Equations

11.1 Linear Algebraic Equations

- Systems of linear equations
- Matrix formulation

11.2 Nonlinear Equations

- Root-finding concepts
- Single nonlinear equations with `fzero`

11.3 Iterative Root-Finding Methods

- Bisection method, Secant method
- Convergence criteria and error estimation
- Comparison of root-finding methods

11.4 Systems of Nonlinear Equations

- Using `fsolve`
- Initial guesses and multiple solutions
- Scaling considerations
- Chemical equilibrium examples

12 Numerical Differentiation and Integration

12.1 Numerical Differentiation

- Numerical vs. symbolic differentiation
- Finite-difference concept
- Forward, backward, and central differences
- Numerical differentiation in MATLAB

12.2 Numerical Integration

- Numerical vs. symbolic integration
- Trapezoidal integration using `trapz`
- Advanced integration with `integral`
- Integration of experimental data

12.3 Accuracy and Error

- Step size and numerical error
- Accuracy considerations

13 Interpolation, Curve Fitting, and Basic Regression

13.1 Interpolation

- Interpolation concepts
- Linear interpolation using `interp1`
- Extrapolation

13.2 Curve Fitting

- Polynomial fitting and basic regression
- Comparing fitted and experimental data

13.3 Statistical Evaluation

- Residuals and coefficient of determination (R^2)
- Error analysis
- Model quality assessment

14 Basic Debugging and Efficient MATLAB Coding

14.1 Debugging

- Understanding error messages
- Syntax, runtime, and logical errors
- Common MATLAB mistakes

14.2 Debugging Tools

- Breakpoints and step-by-step execution
- Inspecting variables
- Command Window debugging

14.3 Efficient Coding

- Vectorization techniques
- Avoiding unnecessary calculations
- Basic preallocation
- Efficient matrix operations
- Readable and maintainable code

14.4 Verification

- Checking intermediate results
- Dimensional and physical consistency
- Comparing numerical and analytical results

15 MATLAB with VS Code and AI-Assisted Coding

15.1 MATLAB in VS Code

- MATLAB extensions and syntax highlighting
- Running MATLAB code from VS Code
- Working with scripts and functions

15.2 Using AI for MATLAB Programming

- Using AI to explain MATLAB syntax
- Generating MATLAB code with AI
- Converting algorithms into MATLAB
- Debugging with AI
- Improving and optimizing code

15.3 Responsible AI-Assisted Programming

- Verifying generated code
- Identifying hallucinated functions
- Checking numerical correctness
- Testing AI-generated code
- Data and privacy considerations
- AI as a programming assistant rather than a substitute

16 Introduction to Chemical Engineering MATLAB Projects

16.1 Problem Formulation

- Translating engineering problems into MATLAB
- Defining inputs, outputs, and assumptions

16.2 Implementation

- Breaking problems into functions
- Calculation workflow
- Data input, computation, and visualization

16.3 Verification and Presentation

- Checking physical plausibility
- Comparing with analytical/known results
- Error analysis
- Presenting MATLAB results

16.4 Introductory Projects

- Material balance calculations
- Energy balance computations
- Reaction kinetics analysis
- Experimental data analysis
- Basic reactor calculations